Business Plan

“OptiMindAI - It is an integration AI platform for people with disabilities.”

ITMLab 2025

Date: 01.09.2025

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Executive Summary

Company Overview

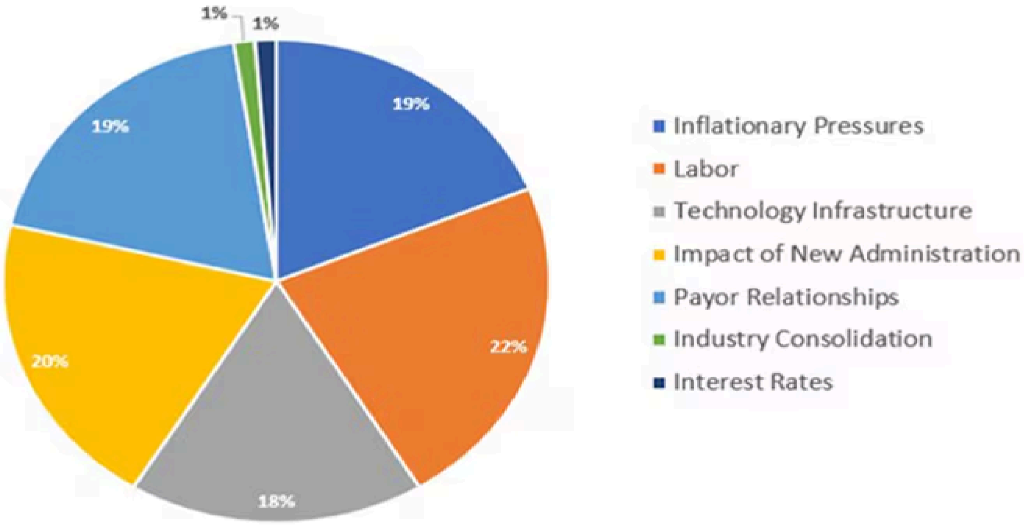
OptiMindAI is an AI-powered healthcare assistant platform that integrates advanced computer vision, natural language processing, and predictive analytics to support patients, healthcare providers, and institutions. Our mission is to make healthcare more accessible, efficient, and inclusive by providing intelligent digital tools for health monitoring, disease prevention, and patient engagement.

The project originated as a research and innovation initiative and has grown into a scalable platform with strong potential for commercialization in both emerging and developed markets.

Market Opportunity

The global healthcare industry faces critical challenges:

- Increasing demand for healthcare services due to aging populations and rising chronic diseases.
- Limited accessibility to qualified medical professionals in underserved regions.
- High operational inefficiencies within hospitals and clinics.
- Growing costs and pressure on insurance systems.



The AI in healthcare market is projected to grow from $20 billion in 2024 to $187 billion by 2030, with an average annual growth rate exceeding 35%. OptiMindAI is strategically positioned to capture this opportunity by targeting:

- Clinics & hospitals seeking to reduce workload and improve diagnostics.
- Insurance companies aiming to minimize fraud and optimize claims.
- Governments & NGOs investing in accessible healthcare solutions.
- End-users (patients) in need of personal health monitoring and AI-driven consultations.

Product Overview

OptiMindAI offers a modular digital health platform with the following core features:

- AI-powered health monitoring: Computer vision for detecting patient conditions, facial recognition for stress/fatigue, and anomaly tracking.
- Virtual health assistant: Natural language interaction enabling medical Q&A, reminders, and guidance.
- Accessibility features: Inclusive solutions tailored for people with disabilities, including voice interaction, gesture recognition, and adaptive UI.
- Predictive analytics: Early risk detection and health trend analysis based on user data.

The product is designed to be user-friendly, scalable, and secure, ensuring compliance with healthcare data protection standards (GDPR, HIPAA).

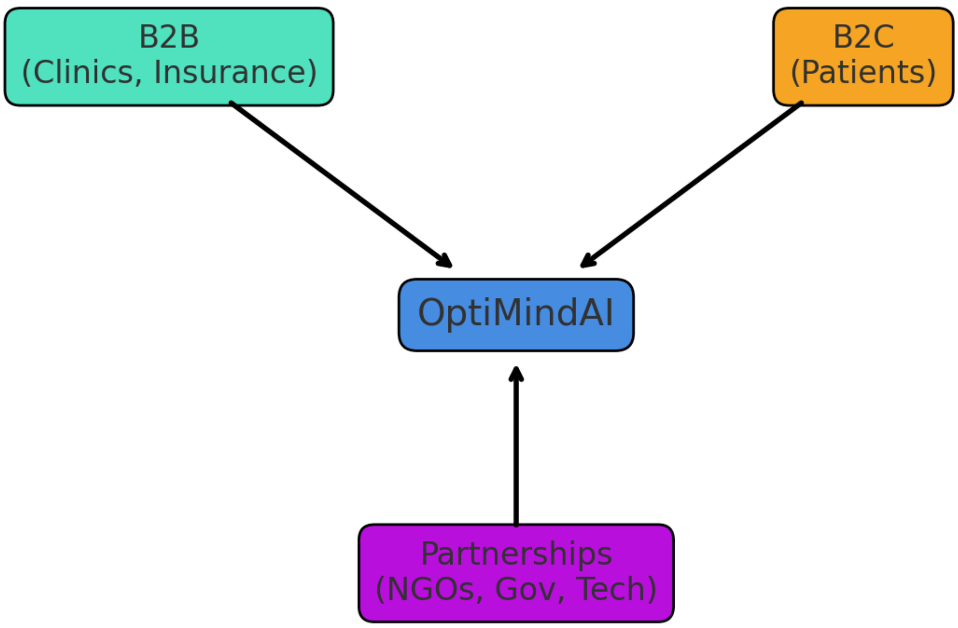
Competitive Advantages

- Accessibility-first approach: Unlike many competitors, OptiMindAI prioritizes inclusivity for vulnerable populations.
- Integrated technology stack: Combination of vision, speech, and predictive AI models in a single platform.
- Research-backed foundation: Developed with input from AI researchers and healthcare professionals.
- Scalable monetization model: B2B partnerships with clinics/insurance companies and B2C subscriptions for individual users.

Business Model

OptiMindAI will generate revenue through:

1. B2B subscription licenses for clinics, hospitals, and insurance companies.
2. B2C freemium model with premium AI features for individual users.
3. Custom integrations for governments and large institutions.
4. Partnerships with NGOs for deployment in underserved regions.



Financial Highlights

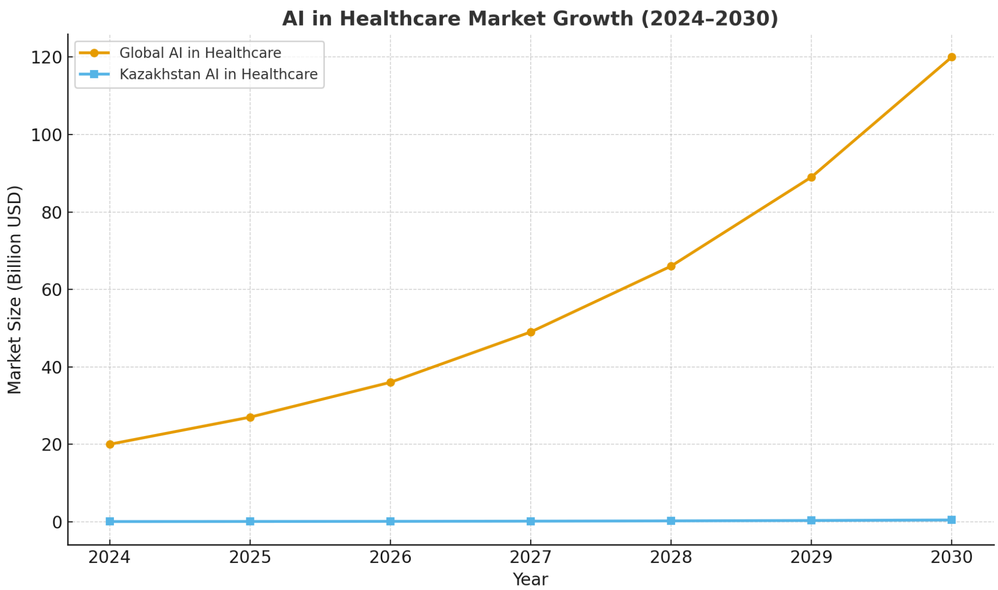
- 6 months: MVP release, pilot programs with 2–3 clinics, initial B2C adoption (~$50K in revenue).
- 12 months: Expansion to 10+ clinics, insurance company partnerships, reaching ~$250K in revenue.
- 24 months: International expansion (CIS, Europe, Asia-Pacific), projected revenue ~$1.2M with positive EBITDA.

Investment will be primarily allocated to:

- Product development & AI model enhancement (40%).
- Clinical validation & regulatory compliance (25%).
- Marketing & market entry strategies (20%).
- Operations & team expansion (15%).

Funding Needs

We are seeking an initial investment of $500,000 to accelerate development, secure clinical partnerships, and expand market presence. This funding will ensure rapid scaling and prepare the platform for global expansion.



Vision & Impact

OptiMindAI is not only a commercial venture but also a socially significant project. By addressing healthcare accessibility gaps, particularly for people with disabilities and underserved populations, we aim to:

- Improve patient quality of life.
- Support medical professionals in reducing workload.
- Contribute to sustainable healthcare systems worldwide.

Our long-term vision is to become a leading AI health assistant globally, integrating seamlessly into both personal and institutional healthcare ecosystems.

2. Information on the Current Activities of the Project Initiator

The initiator of the project is a multidisciplinary team of young professionals specializing in Artificial Intelligence, Computer Vision, and Human–Computer Interaction, with a proven track record of developing assistive technologies. The team has already implemented several working prototypes that form the technological foundation of OptiMindAI.

Current Achievements

- AI-Powered Vision Tools: Implemented real-time object detection (YOLOv5) and face recognition (MTCNN + face_recognition) modules.
- Interactive Map Integration: Developed and integrated a district-level navigation system using Folium + GeoJSON, enabling geospatial support for accessibility solutions.
- Voice Assistant: Built a functional speech recognition and synthesis module using `speech_recognition` and `pyttsx3`, with a command interface for intuitive user interaction.

- Unified Interface: Combined vision, voice, and mapping capabilities into a single Tkinter-based application, serving as the initial MVP.

Research & Validation

- Conducted user testing sessions with representatives from local disability organizations to identify pain points in navigation, verification, and daily assistance.

- Collaborated with experts in healthcare technology and inclusive design to ensure the solution meets accessibility standards.

Community Engagement

- Active cooperation with disability associations in Almaty and across Kazakhstan to align the product with real user needs.

- Participation and victory in hackathons focused on AI for accessibility, showcasing both innovation and execution capabilities.

Strategic Readiness

The initiator is currently engaged in:

- Scaling the MVP to include sign language translation, smart routing, and IoT integration (smart cane, glasses, and home cameras).

- Establishing partnerships with universities, clinics, and NGOs to pilot the technology.

- Preparing the platform for integration with e-government and healthcare services.

The combination of technical expertise, community validation, and early achievements provides a solid foundation for the successful implementation and scaling of OptiMindAI.

3. Project Description

OptiMindAI is an inclusive AI-driven platform designed to support people with disabilities by integrating multiple assistive technologies into one unified system. Unlike existing fragmented solutions that address individual challenges in isolation, OptiMindAI provides a centralized and adaptive ecosystem that combines computer vision, speech technologies, geospatial mapping, and IoT integration.

Core Idea

The project aims to remove everyday barriers faced by people with visual, hearing, and mobility impairments by offering an intelligent assistant that ensures independence, safety, and equal access to information and services.

Key Functionalities

- Computer Vision Modules: Real-time object detection, obstacle recognition, and face identification for navigation and verification.

- Voice Assistant: Speech recognition and synthesis to enable natural, hands-free interaction.

- Sign Language Translator: AI-based gesture recognition for translation between sign language and spoken text.

- Smart Routing: Personalized navigation with awareness of urban barriers, integrating maps and cameras.

- IoT Integration: Support for smart canes, glasses, and home AI cameras to enhance accessibility indoors and outdoors.

- E-Government & Healthcare Access: Secure authentication and assistance for people with visual impairments in official digital services.

Innovation

OptiMindAI differs from conventional assistive tools by offering a modular, scalable, and cross-disability solution. While most existing products are limited to single-use cases (e.g., screen readers, white canes, or translation apps), our platform consolidates multiple capabilities into one accessible ecosystem, reducing fragmentation and improving usability.

Target Impact

The solution addresses a wide range of challenges highlighted in disability studies, including mobility support, digital inclusion, communication barriers, and independent living. With 737,500 people with disabilities in Kazakhstan alone, the project has strong potential for local impact and international scalability.

Mission

Our mission is to empower people with disabilities through AI, creating tools that extend human capability, improve autonomy, and foster inclusion in society.

OptiMindAI is an AI-powered assistive technology platform designed to enhance independence and accessibility for people with disabilities. The system integrates computer vision, speech recognition, natural language processing, and IoT connectivity into a unified ecosystem that adapts to both personal and institutional use cases.

Core Components

1. AI Vision Module

    - Object recognition (YOLOv5)

    - Face detection and identification (MTCNN + Face_Recognition)

    - Gesture recognition and sign language interpretation (MediaPipe + custom-trained models)

- Smart navigation: detection of obstacles, environmental barriers, and safe routing

2. Voice Assistant Module
   - Speech recognition (SpeechRecognition API, Google STT)
   - Voice output and text-to-speech (pyttsx3, multilingual synthesis)
   - Command execution for device interaction (mobile app, smart glasses, IoT devices)
   - Conversational AI layer for natural communication
3. Adaptive Interfaces
   - Mobile application (Flutter)
   - Smart glasses integration (voice + AR overlays)
   - Smart cane with sensors and audio feedback
   - IoT-enabled home assistant cameras with embedded recognition models
4. Cloud & Data Layer
   - Centralized database for personalization and user preferences
   - Secure API for integration with government portals, clinics, and partner apps
   - Scalable architecture enabling both B2B (institutions) and B2C (end-users) deployment

Key Financial Metrics

1. Customer Acquisition Cost (CAC)

Customer Acquisition Cost represents the total marketing and sales expenses required to acquire one new customer.

- In the Kazakhstan market, we project an average CAC of 5,000 KZT per user during the first year.
- This includes digital advertising, awareness campaigns, and onboarding support.
- Benchmark: Sustainable SaaS and health-tech projects typically aim for CAC < 10,000 KZT in emerging markets.

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2. Average Revenue per User (ARPU)

Average Revenue per User is the monthly income generated from each paying customer.

- Our subscription model is priced at 3,000 KZT per month.
- Therefore, ARPU = 3,000 KZT.
- On an annualized basis, this equals 36,000 KZT per user per year.

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3. Lifetime Value (LTV)

Lifetime Value measures the total net revenue expected from a single customer across their entire engagement with the product.

- Based on projected churn and retention rates, we assume an average customer lifetime of 24 months.

- LTV = ARPU × Customer Lifetime
- LTV = 3,000 KZT × 24 = 72,000 KZT per customer.

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4. LTV/CAC Ratio

This ratio is a critical metric for evaluating business sustainability.

- With LTV = 72,000 KZT and CAC = 5,000 KZT, our LTV/CAC = 14.4.
- A ratio above 3 is considered strong in SaaS and health-tech markets. Our current projection indicates high efficiency and profitability in scaling.